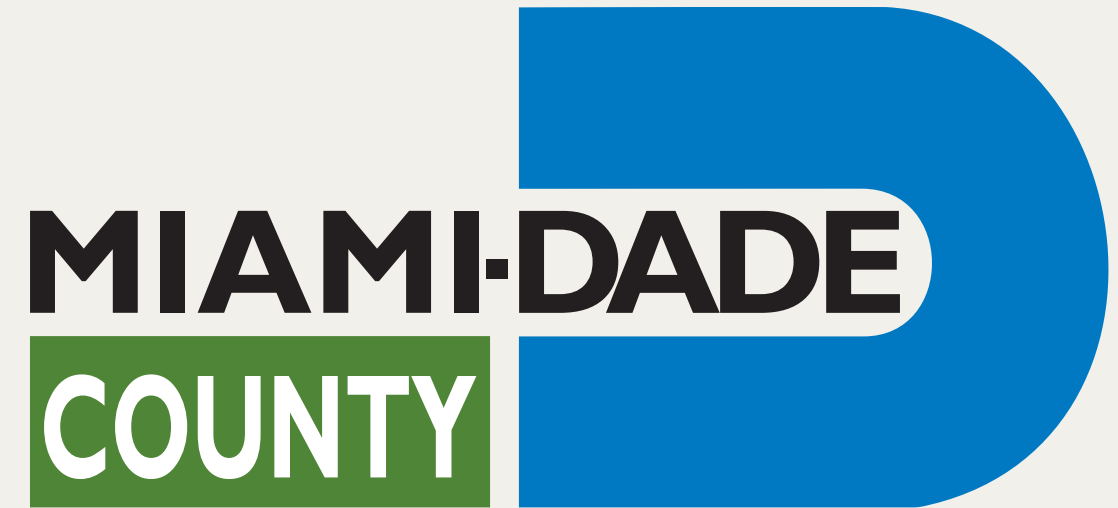


**From Orders to
Opportunities:
Strategic Insights for
Miami-Dade County
Suppliers**



DATA MINING PROJECT OVERVIEW – “MIAMI-DADE COUNTY 2024 PURCHASE ORDERS”

PROJECT REQUIREMENTS

Project Objective: Use Data Mining techniques on a Miami-Dade County dataset. Also incorporate an app that provides a real-world solution based on the dataset findings.

Required Tools:

- ✓ Google Colab/Python: Extract, transform, load, analyze, and visualize data
- ✓ Github: Online repository reference point for the project
- ✓ Jira Project Management: Manage project deadlines
- ✓ Power Point: Presentation creation
- ✓ Windsurf: Application creation

Note: The presentation that follows this slide is the actual presentation. Once the presentation is over, there are slides showing the application that is associated with this project. For details regarding this project, please visit my portfolio website located in my LinkedIn bio and click on “Reverse RFM Analysis on Miami-Dade Purchase Orders.” Links to the dataset, Github repo, and application are in the “Appendix” section of this presentation.

PROJECT DETAILS

Dataset: Miami-Dade County 2024 Purchase Orders

Goal: Mine the dataset for insights that can potentially provide “purchasing power” negotiation points with suppliers for better pricing opportunities.

Techniques:

- ✓ “Reverse” RFM Analysis: Understanding the recency of purchases, frequency of purchases, and monetary value of purchases from suppliers
- ✓ K Means Clustering: Use the RFM data to segment purchases into clusters as potential negotiation points specifically for each cluster
- ✓ Monte Carlo Analysis: Forecasting Miami-Dade County’s potential spending with each supplier

Details: RFM methodologies are typically used by suppliers to understand consumer behavior. In this project, we will explore how we can use “reverse” RFM methodologies from the business consumer point of view to determine consumer worth to a supplier. We will then use these insights as potential negotiating tools with suppliers to discuss better pricing opportunities.

From Orders to Opportunities: Strategic Insights for Miami-Dade County Suppliers

Presented by David Perez

Presented on April 30, 2025



THE AGENDA

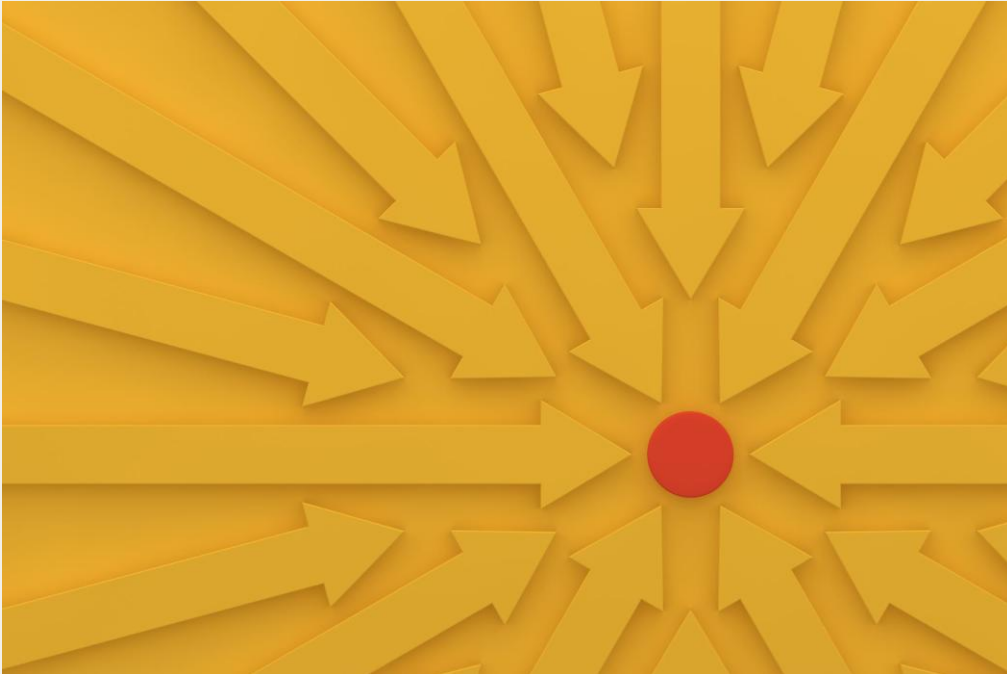


- ≡ Introducing the objective
- 🎯 The methodology of attaining the objective
- 🔍 Analysis and results
- 📊 Case study
- 📋 Recommendations
- 📖 Conclusion
- 📄 Appendix

An aerial photograph of a coastal city at sunset. The sun is low on the horizon to the left, casting a golden glow over the water and sky. The city skyline, featuring numerous skyscrapers, is visible on the right side of the frame. A bridge spans the water in the middle ground. The word "INTRODUCTION" is overlaid in large, white, sans-serif capital letters across the center of the image.

INTRODUCTION

THE OBJECTIVE



Identifying opportunities for Miami-Dade County to utilize its significant purchasing power to negotiate more favorable agreements with its suppliers.

PURCHASING POWER



In 2024, Miami-Dade County allocated more than \$4B in expenses. Given such significant purchasing power, there must be opportunities to strategically leverage this financial capacity.

2024 PURCHASE ORDER SUMMARY



- ✓ 43,058 total transactions
- ✓ 31,069 unique items
- ✓ 1,383 various suppliers
- ✓ Total expenditure of \$4,288,751,653.82



METHODOLOGY

IDENTIFYING OPPORTUNITY



We will be using the Miami-Dade County 2024 Purchase Order dataset to conduct a “Reverse” RFM Analysis and Monte Carlo Analysis. We will be also demonstrating a case study to show this methodology in action.

WHAT IS AN RFM ANALYSIS?



An RFM Analysis evaluates transactions based on three dimensions: Recency (how recently a transaction occurred), Frequency (how often transactions take place), and Monetary Value (the total monetary number of transactions).

WHAT IS A MONTE CARLO ANALYSIS?



A Monte Carlo Analysis is a multiple random simulation technique employed to assess the likelihood of various potential outcomes that assist in risk evaluation and informed decision making.

USING RFM AND MONTE CARLO



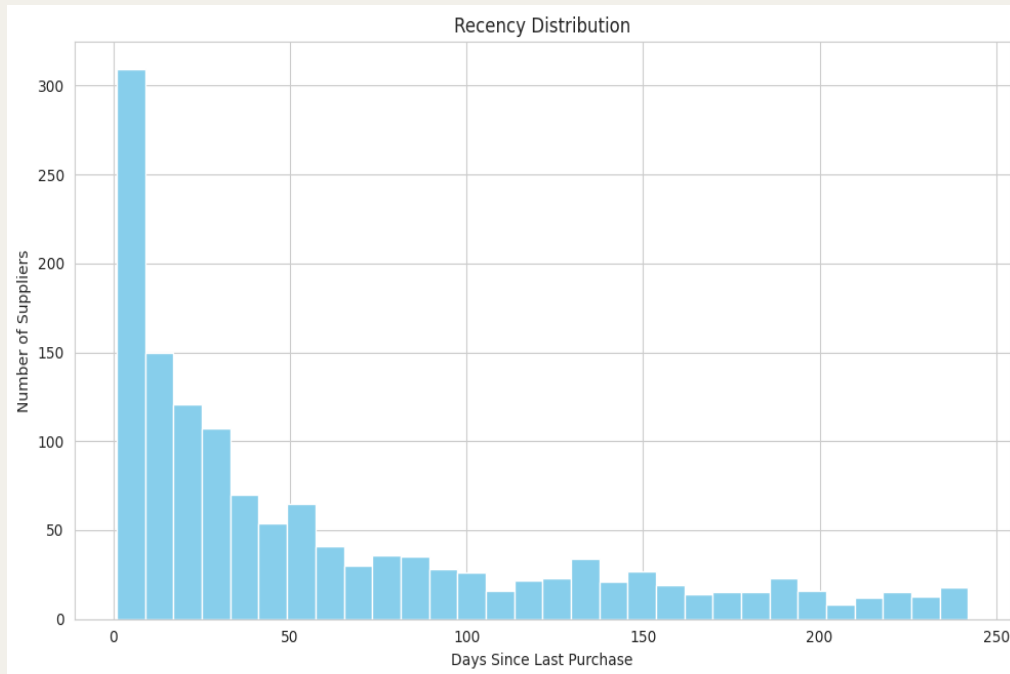
We employed RFM analysis to assess the recency, frequency, and monetary value of purchases made by Miami-Dade County for each supplier.

The Monte Carlo Analysis provides an estimation of Miami-Dade County's projected value to suppliers by forecasting its spending in 2025.

A vibrant fairground scene featuring a carousel on the left, a zip line with people in the center, and a 'DAIQU' booth on the right. The booth has a large sign that says 'DAIQU' in colorful letters, with 'FROZEN' above it and 'NON-ALCOHOLIC' below it. The scene is filled with people and festive decorations.

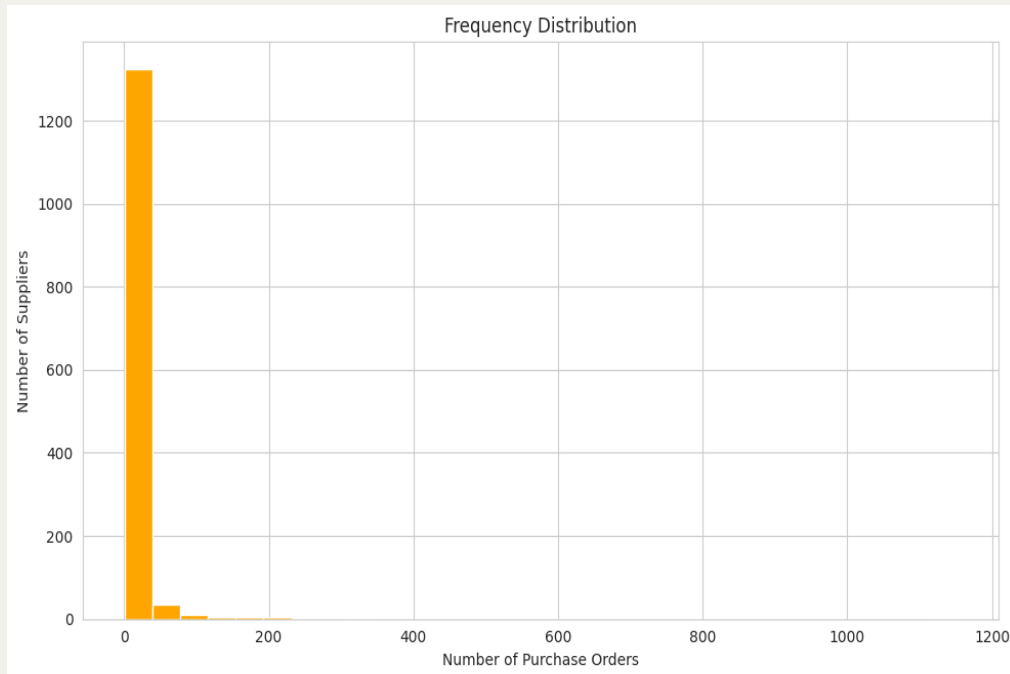
ANALYSIS

THE RECENCY



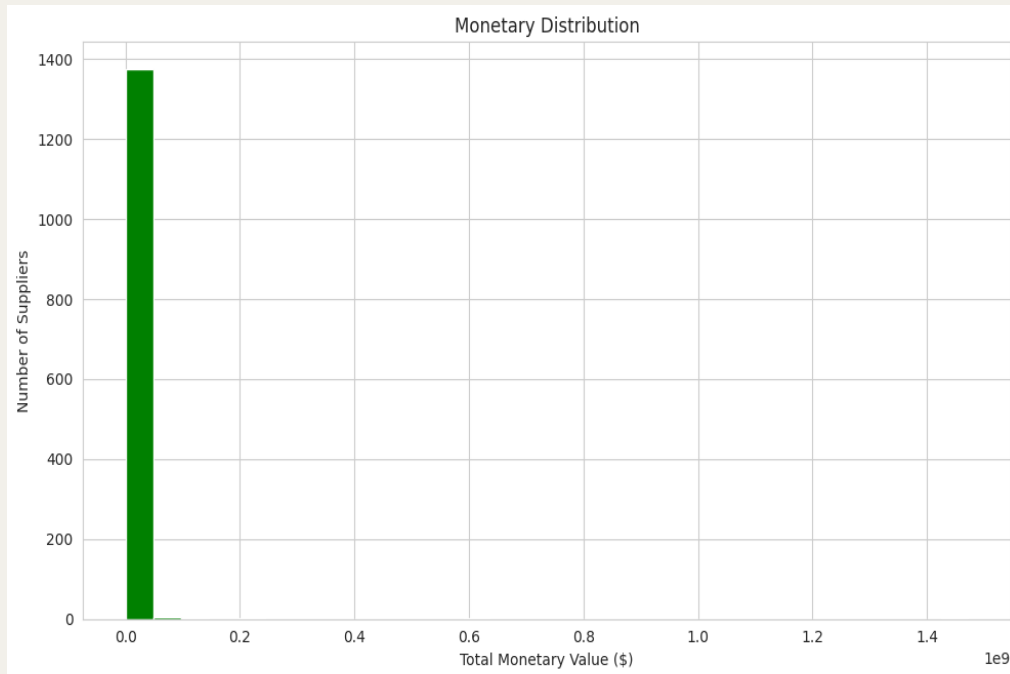
The recency distribution indicates a downward trend, suggesting that a significant quantity of items have been acquired in the recently. This implies that there may be some negotiating power regarding recency when engaging with suppliers.

THE FREQUENCY



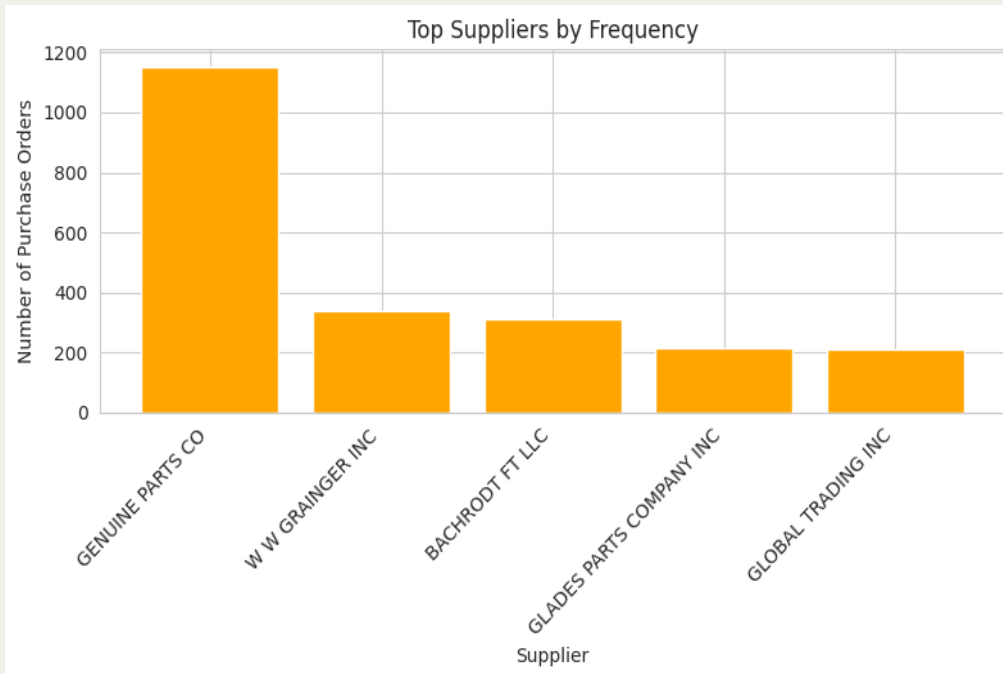
The frequency distribution shows us that a significant volume of purchases is directed towards the same suppliers. This indicates potential leverage in frequency when engaging in negotiations with suppliers.

THE MONETARY



The monetary distribution shows us that most suppliers are being compensated at roughly equal levels. This suggests a potential avenue to enhance cost efficiency during negotiations with these suppliers.

SUPPLIER EVALUATION



We identified the primary suppliers from whom Miami-Dade County made the most frequent purchases, those with the highest monetary value, and the suppliers with the most recent transactions. To illustrate how this data can be beneficial for business negotiations, we chose to evaluate Genuine Parts Co., the supplier with the highest purchase frequency, as a case study.

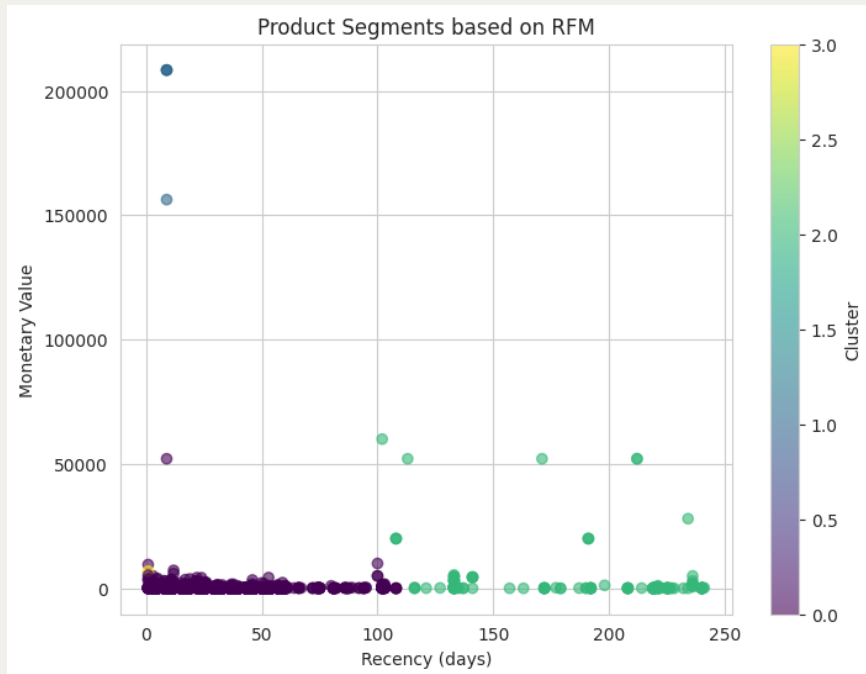
CASE STUDY

GENUINE PARTS CO



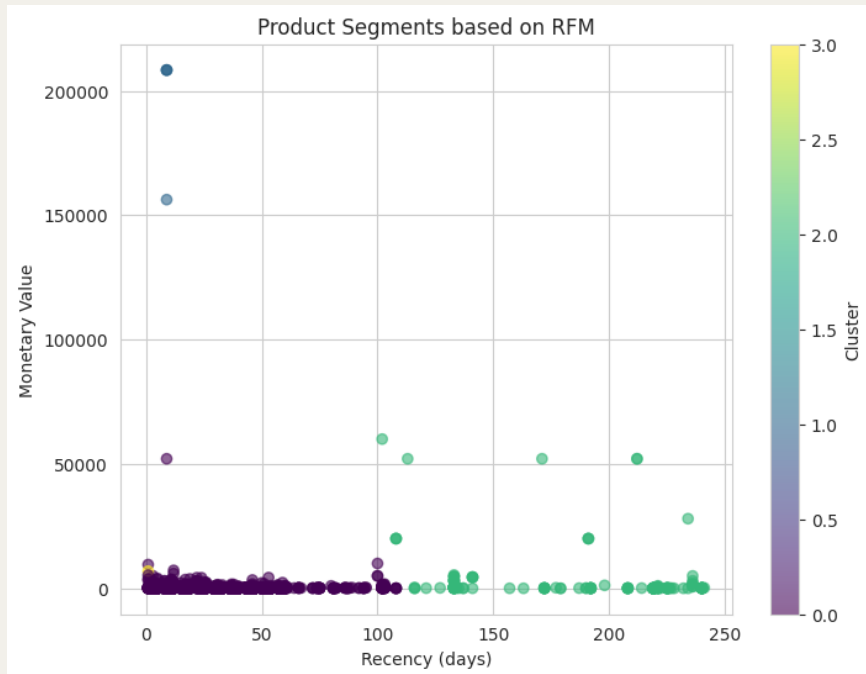
Miami-Dade County
expended \$876,747.18,
acquiring 2,276 distinct
items from Genuine
Parts Co in 2024.

PRODUCT SEGMENTATION



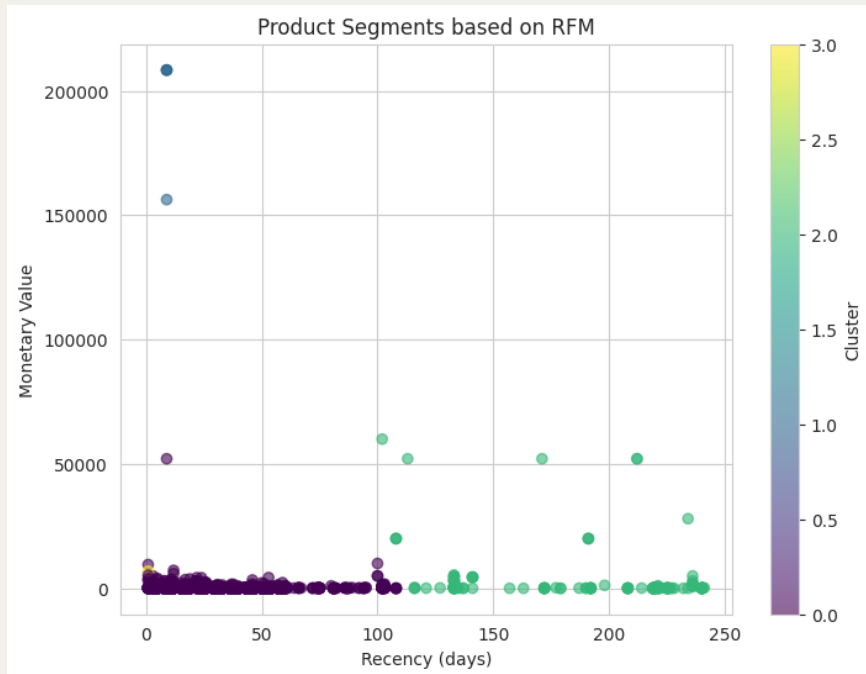
We employed a clustering algorithm based on our RFM Analysis to categorize products from Genuine Parts Co into distinct groups.

SEGMENTATED CLUSTER 0



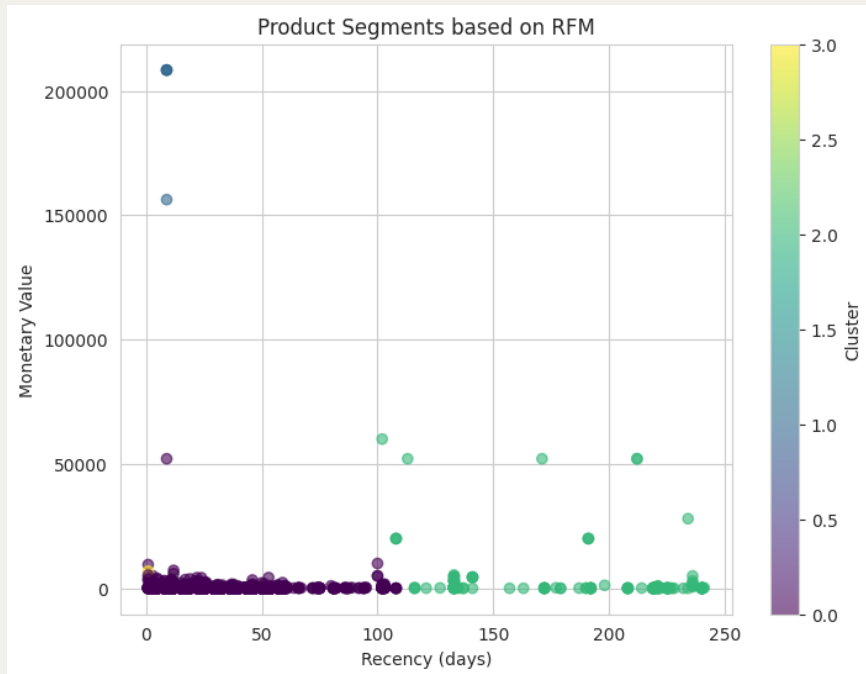
The Cluster 0 group (Purple) contains a total of 2,132 items, characterized by a low monetary value per item overall, yet these items are frequently and recently purchased.

SEGMENTATED CLUSTER 1



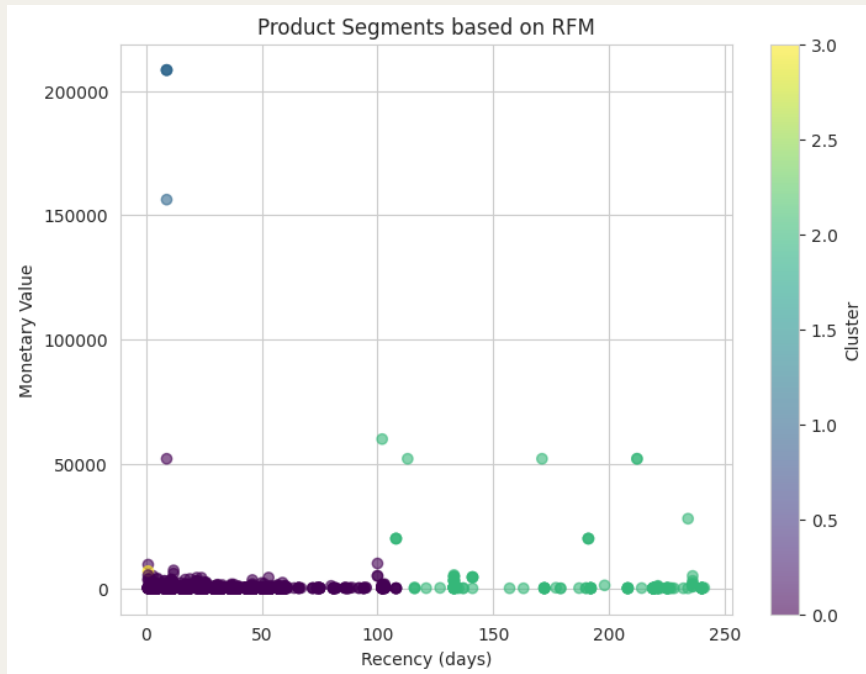
The Cluster 1 group (Blue) includes 4 items, representing unusually large purchases that are not made very often.

SEGMENTATED CLUSTER 2



The Cluster 2 group (Green) has 148 items, which are infrequently purchased and have the second lowest monetary value among the four clusters.

SEGMENTATED CLUSTER 3



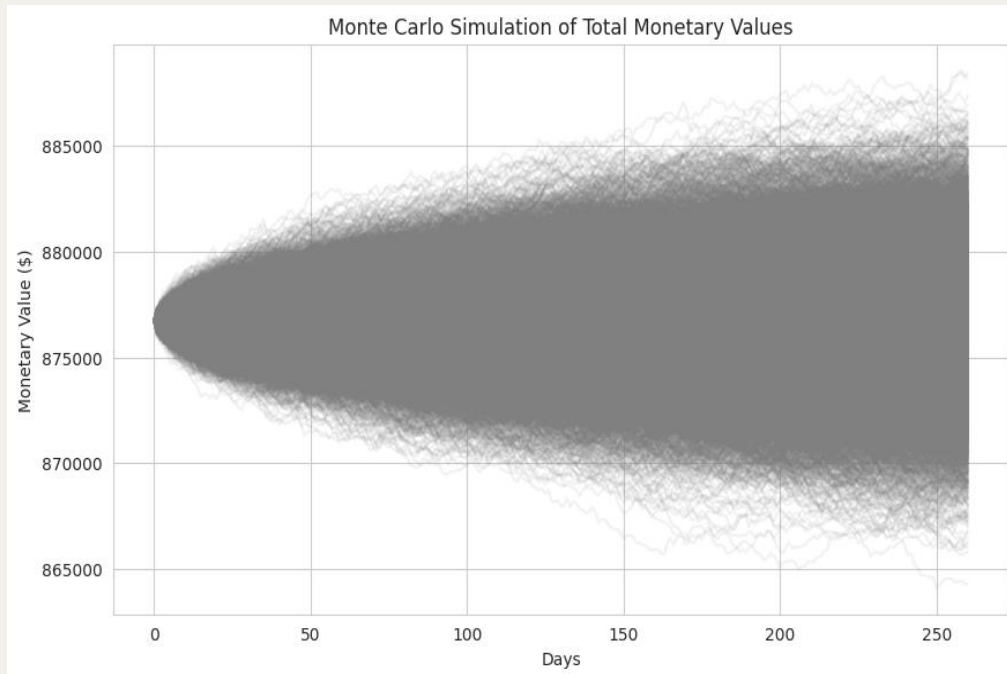
The Cluster 3 group (Yellow) consists of a single core item, which has been purchased 66 times and was acquired recently.

SEGMENTATION CONCLUSION



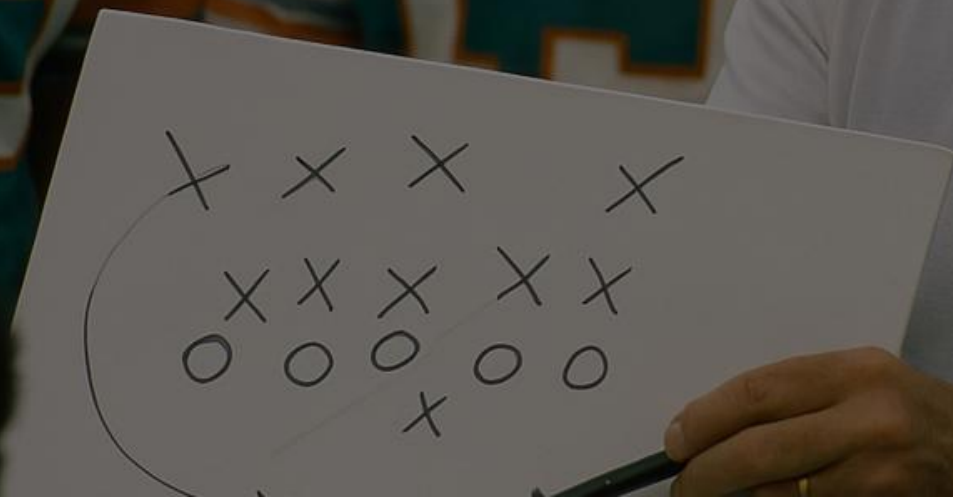
Clusters 0 and 3 demonstrate the highest potential, with Cluster 0 exhibiting a total monetary value of over \$572K and Cluster 3 showing a total of almost \$7k. The items within these clusters are frequently purchased and been acquired recently, making them ideal candidates for negotiation discussions with the supplier.

MONTE CARLO ANALYSIS



After conducting 10,000 simulations with a 95% confidence interval over a business year, Miami-Dade County's forecasted value to GPC in 2025 is \$875K with any variation falling between \$872K and \$877K. We will use this data to highlight potential losses if Miami-Dade County decides to shift their business elsewhere.

RECOMMENDATIONS

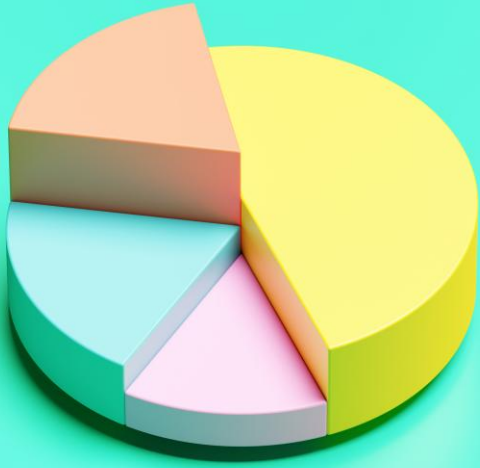


DISCOUNTED RATES



- ✓ Miami-Dade County is eligible to seek an overall discount on the anticipated expenditure of \$875K with Genuine Parts Co in 2025.
- ✓ Miami-Dade County has the option to request that discounts be structured in tiers. For example, a 10% discount will be applied upon exceeding the \$100k mark, followed by a 15% discount once the \$300k threshold is reached, and so on.

SEGMENTED DISCOUNTS



- ✓ Cluster 0 holds a total monetary value of over \$572K. Given its substantial value, it may be feasible to apply an overall discount to all items within this cluster.
- ✓ Cluster 3 has a monetary value of almost \$7K, attributed to a single core item that was acquired 66 times in 2024. This item could be a candidate for a discount, considering its frequent purchases.

NEGOTIATE FOR SPECIFIC ITEMS



Negotiate for items that exhibit a strong combination of frequency and recency. For example, if an item's frequency is 12 or greater, and has a recency threshold of 30 days or fewer, we can infer that these items are likely purchased monthly. With this data, Miami-Dade County can advocate for monthly discounts on these products.

CONCLUSION

A nighttime photograph of a city skyline. In the foreground, a large bridge with multiple arches spans a body of water. The bridge is illuminated with warm yellow lights, and its reflection is visible in the water. In the background, several tall skyscrapers are lit up with various colors, including yellow, blue, and red. The sky is dark with some clouds. The word "CONCLUSION" is overlaid in the center of the image in a large, white, sans-serif font.

THE SUMMARY



This comprehensive analysis can be applied to any supplier within the dataset and serves as a valuable negotiating tool for securing improved pricing. Achieving better pricing could lead to Miami-Dade County saving potentially hundreds of thousands of dollars, or even more based on negotiations, which can then be redirected towards other county initiatives. Furthermore, this data may provide leverage in discussions with rival suppliers to obtain more favorable pricing if current suppliers are unwilling to negotiate.

APPENDIX



FURTHER ANALYSIS



- ✓ Access to completed datasets from prior years can enhance granularity, enabling more precise RFM and Monte Carlo predictions.
- ✓ Competitor price sheets would also be beneficial during negotiations.
- ✓ Any contracts or agreements currently in use with suppliers.
- ✓ Incorporating categorical data could further bolster the negotiation strategy.
- ✓ Collaborating with specialists in each supplier domain or developing a detailed guide on supplier business relations for various sectors would also prove advantageous.

LINKS

Application Link: <https://miami-dade-po-analysis.netlify.app/>

Github Link: https://github.com/Perez786/DataMining_FinalProject

Dataset Link: https://gis-mdc.opendata.arcgis.com/datasets/673f9be756ce4d2a95039bf16d3a3f7d_0/explore

The presentation part of the project has concluded. Now I will take you through the application and briefly show visualizations associated with the Jira and Github aspects of the project.

Application Introduction

The purpose of the application is to give you instant insights regarding Miami-Dade County's relationship with its suppliers. How much money is spent, how often, how recent, which items are being purchased, etc. This can be a useful tool that can expediate lengthy coding that would need to be applied to each individual supplier using traditional analytical methodology when exploring potential negotiation strategies. This React app was built with the assistance of Windsurf and deployed on Netlify.

Application Landing Page

The landing page for the app takes you to a filtered version of the dataset with the columns that are applicable to RFM analysis as well as a description of the items as a reference.

Miami-Dade Purchase Orders 2024

Filter by Supplier

All Suppliers

ID	Supplier Name	PO Date	PO Number	PO Amount	Item Description
0	APPLAUSE YOUR PAWS INC	2/13/2024	1838	\$11,500	INDIVIDUAL CLASSES Instructor Lead Basic Training Le...
1	HOME ENTERTAINMENT SYSTEMS INC	1/8/2024	1772	\$204	Motorola Multi-Unit Charger - Model No. 56531 - 6 Slot St...
2	GARBER FORD INC	5/1/2024	1962	\$313,968.4	Tracking: ASD 2 23/24 2024 (or Newer) Ford Transit T-15...
3	AGNI ENTERPRISES LLC	3/27/2024	1910	\$27	Economy PVC Water Boots - Color: Black - Size 4
4	ZOETIS US LLC	5/7/2024	1971	\$2,335.5	Canine and Feline Serum Luteinizing Hormone (LH) Test
5	INTERBORO PACKAGING CORP	5/15/2024	1978	\$1,581.6	40X46 3.00 mil 50/Case Red Liners - (40-45)Gallon Item ...
6	GLOBAL COURIER SERVICES INC	7/16/2024	2079	\$3,000	Deliveries/Pickups from & to Vet Stations
7	INTERBORO PACKAGING CORP	5/15/2024	1978	\$618.6	Trash Bag Liner - Color: RED - 55 to 60 Gal - Estimated ...
8	INTERBORO PACKAGING CORP	5/15/2024	1978	\$618.6	Trash Bag Liner - Color: RED - 55 to 60 Gal - Estimated ...
9	INTERBORO PACKAGING CORP	5/15/2024	1978	\$618.6	Trash Bag Liner - Color: RED - 55 to 60 Gal - Estimated ...

Rows per page: 100 1-100 of 42763

Application Functionality

From the landing page, you can click on the drop-down box located in the top left and click on a supplier to review their RFM analysis.

Miami-Dade Purchase Orders 2024

GARBER FORD INC

GARCIA MULLIN GROUP LLC

GARTNER INC

GASSANT ENTERPRISES LLC

GAYLORD BROS INC

GCS IMAGING INC

GEISMAR NORTH AMERICA INC

GENASYS INC

GENERAL WELDING SERVICE ENTERPRISES INC

GENERATING SYSTEMS INC

GENESIS ELECTRICAL CORP

GENUINE PARTS CO

GEOSOL INC

GEOTYPE OF SOUTH FLORIDA INC

GHD SERVICES INC

GHX INDUSTRIAL LLC

GILLIG LLC

GL DISTRIBUTORS INC

GLADES PARTS COMPANY INC

	PO Number	PO Amount	Item Description
24	1838	\$11,500	INDIVIDUAL CLASSES Instructor Lead Basic Training Le...
4	1772	\$204	Motorola Multi-Unit Charger - Model No. 56531 - 6 Slot St...
4	1962	\$313,968.4	Tracking: ASD 2 23/24 2024 (or Newer) Ford Transit T-15...
24	1910	\$27	Economy PVC Water Boots - Color: Black - Size 4
4	1971	\$2,335.5	Canine and Feline Serum Luteinizing Hormone (LH) Test
24	1978	\$1,581.6	40X46 3.00 mil 50/Case Red Liners - (40-45)Gallon Item ...
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24	1978	\$618.6	Trash Bag Liner - Color: RED - 55 to 60 Gal - Estimated ...

Rows per page: 100 ▾ 1-100 of 42763 < >

Application Results

Once a supplier is selected, you can view their RFM data as well as when Miami-Dade County first purchased an item from them in 2024, the last time they ordered, and the total amount of orders. Note: the recency value shows up to the current date, so the recency value will increase as the days go by without a new purchase.

Miami-Dade Purchase Orders 2024

Filter by Supplier

GENUINE PARTS CO

RFM Analysis for GENUINE PARTS CO

Recency

251 days

Days since last purchase order

Frequency

370.02 orders

Average orders per month

Monetary

\$876,747.18

Average order value: \$299.333

Order History:

First Order: 1/2/2024

Last Order: 8/30/2024

Total Orders: 2929

ID	Supplier Name	PO Date	PO Number	PO Amount	Item Description
0	APPLAUSE YOUR PAWS INC	2/13/2024	1838	\$11,500	INDIVIDUAL CLASSES Instructor Lead Basic Training Le...
1	HOME ENTERTAINMENT SYSTEMS INC	1/8/2024	1772	\$204	Motorola Multi-Unit Charger - Model No. 56531 - 6 Slot St...
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